

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1300	0.0390	3.33 $\times$	0.1594	0.1857	0.86 $\times$
DBQuery	15	13	0.1223	0.0883	1.39 $\times$	0.1375	0.2008	0.68 $\times$
DecisionTree	7	6	3.872	3.151	1.23 $\times$	–	–	–
DomTree	15	14	0.1403	0.0623	2.25 $\times$	0.0585	0.0885	0.66 $\times$
KDTree	17	16	0.1833	0.1410	1.30 $\times$	–	–	–
LinearListReduction	11	11	0.0286	0.0050	5.74 $\times$	–	–	–
List	2	2	0.0880	0.0903	0.97 $\times$	0.2961	0.3338	0.89 $\times$
MonoTree	3	2	0.0236	0.0225	1.05 $\times$	0.0463	0.0421	1.10 $\times$
ObjectGraph	5	4	0.1033	0.0807	1.28 $\times$	0.1505	0.1565	0.96 $\times$
PiecewiseFunctions	8	7	0.1408	0.0909	1.55 $\times$	0.2395	0.2690	0.89 $\times$
TernaryTree	5	3	0.0299	0.0292	1.02 $\times$	0.0843	0.0773	1.09 $\times$
Trie	10	9	0.0635	0.0332	1.91 $\times$	0.1496	0.1882	0.79 $\times$

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. $>1\times$ means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	1.54 $\times$	0.48 $\times$	3.70 $\times$	0.67 $\times$	14.57 $\times$	0.33 $\times$	6.80 $\times$
DBQuery	0.89 $\times$	0.41 $\times$	1.20 $\times$	0.35 $\times$	8.94 $\times$	0.29 $\times$	6.30 $\times$
DecisionTree	1.23 $\times$	0.69 $\times$	18.71 $\times$	3.13 $\times$	0.88 $\times$	3.25 $\times$	36.32 $\times$
DomTree	1.42 $\times$	0.53 $\times$	2.43 $\times$	1.07 $\times$	19.60 $\times$	0.53 $\times$	10.66 $\times$
KDTree	1.30 $\times$	0.81 $\times$	1.18 $\times$	1.01 $\times$	0.38 $\times$	1.02 $\times$	4.71 $\times$
LinearListReduction	5.78 $\times$	0.90 $\times$	80.67 $\times$	18.00 $\times$	262.81 $\times$	17.56 $\times$	11.42 $\times$
List	0.90 $\times$	0.74 $\times$	3.60 $\times$	0.55 $\times$	0.13 $\times$	1.04 $\times$	1.44 $\times$
MonoTree	1.12 $\times$	0.82 $\times$	0.30 $\times$	0.62 $\times$	0.24 $\times$	0.83 $\times$	1.13 $\times$
ObjectGraph	1.10 $\times$	0.74 $\times$	2.23 $\times$	0.74 $\times$	0.53 $\times$	1.23 $\times$	2.67 $\times$
PiecewiseFunctions	1.08 $\times$	0.58 $\times$	2.35 $\times$	1.18 $\times$	1.56 $\times$	0.65 $\times$	4.25 $\times$
TernaryTree	1.14 $\times$	0.78 $\times$	0.33 $\times$	0.51 $\times$	0.25 $\times$	0.86 $\times$	1.41 $\times$
Trie	0.94 $\times$	0.50 $\times$	3.06 $\times$	0.71 $\times$	3.22 $\times$	0.17 $\times$	6.55 $\times$
Geomean	1.30 $\times$	0.65 $\times$	2.63 $\times$	1.04 $\times$	2.02 $\times$	0.90 $\times$	4.80 $\times$

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	9.7e+08/ 6.3e+08	9.9e+08 /2.1e+09	1.3e+07/ 3.4e+06	2.5e+05 /3.7e+05	1.3e+06/ 9.0e+04	1.9e+04 /5.6e+04	5.1e+07/ 7.4e+06
DBQuery	9.5e+08 /1.1e+09	1.1e+09 /2.6e+09	3.9e+06/ 3.3e+06	1.7e+05 /4.8e+05	8.7e+05/ 9.8e+04	1.2e+04 /4.1e+04	2.0e+07/ 3.2e+06
DecisionTree	2.0e+10/ 1.6e+10	4.3e+10 /6.3e+10	5.9e+08/ 3.1e+07	1.7e+05/ 5.5e+04	1.9e+06 /2.1e+06	1.7e+05/ 5.3e+04	1.1e+09/ 3.0e+07
DomTree	8.4e+08/ 5.9e+08	1.2e+09 /2.3e+09	1.2e+07/ 4.8e+06	1.3e+05/ 1.2e+05	1.1e+06/ 5.6e+04	1.5e+04 /2.9e+04	5.7e+07/ 5.3e+06
KDTree	9.3e+08/ 7.2e+08	3.3e+09 /4.1e+09	1.6e+07/ 1.4e+07	1.0e+03/ 1.0e+03	4.2e+04 /1.1e+05	1.0e+03/ 1.0e+03	4.4e+07/ 9.3e+06
LinearListReduction	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	4.7e+06/ 5.8e+04	7.6e+02/ 4.2e+01	1.9e+06/ 7.3e+03	7.6e+02/ 4.3e+01	1.2e+07/ 1.1e+06
List	1.3e+09 /1.4e+09	4.5e+09 /6.1e+09	9.8e+06/ 2.7e+06	3.1e+05 /5.7e+05	2.4e+04 /1.8e+05	1.5e+04/ 1.4e+04	4.9e+07/ 3.4e+07
MonoTree	2.9e+08/ 2.6e+08	9.1e+08 /1.1e+09	1.6e+05 /5.3e+05	4.7e+04 /7.6e+04	3.4e+03 /1.4e+04	4.1e+03 /4.9e+03	1.9e+06/ 1.7e+06
ObjectGraph	9.4e+08/ 8.6e+08	3.0e+09 /4.0e+09	3.0e+06/ 1.3e+06	1.7e+05 /2.4e+05	3.1e+04 /5.7e+04	2.2e+04/ 1.8e+04	2.2e+07/ 8.3e+06
PiecewiseFunctions	1.3e+09/ 1.2e+09	2.9e+09 /5.0e+09	1.2e+07/ 5.3e+06	4.9e+05/ 4.2e+05	1.5e+05/ 9.3e+04	4.0e+04 /6.2e+04	4.4e+07/ 1.0e+07
TernaryTree	4.3e+08/ 3.7e+08	1.2e+09 /1.6e+09	4.8e+05 /1.5e+06	6.6e+04 /1.3e+05	5.1e+03 /2.0e+04	3.7e+03 /4.4e+03	3.2e+06/ 2.3e+06
Trie	6.5e+08 /6.9e+08	1.2e+09 /2.5e+09	8.6e+06/ 2.8e+06	2.2e+05 /3.0e+05	1.5e+05/ 4.7e+04	7.8e+03 /4.7e+04	2.6e+07/ 3.9e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.2894	1.226	0.2247	<i>OOM</i>
DBQuery	0.2598	0.3344	0.2891	1.468
DecisionTree	3.872	5.816	3.151	28.39
DomTree	0.1988	0.3437	0.1508	1.026
KDTree	0.1833	0.2519	0.1410	0.8018
LinearListReduction	0.0286	0.1628	0.0050	<i>OOM</i>
List	0.3840	<i>OOM</i>	0.4241	<i>OOM</i>
MonoTree	0.0699	0.1507	0.0645	0.2933
ObjectGraph	0.2538	0.5022	0.2371	0.7727
OctTree	0.5791	0.6525	–	<i>OOM</i>
PiecewiseFunctions	0.3803	0.5810	0.3598	2.419
TernaryTree	0.1142	0.1490	0.1065	0.6170
Trie	0.2130	0.3027	0.2214	1.102

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.29 $\times$	4.24 $\times$	5.46 $\times$	–
DBQuery	0.90 $\times$	1.29 $\times$	1.16 $\times$	0.23 $\times$
DecisionTree	1.23 $\times$	1.50 $\times$	1.85 $\times$	0.20 $\times$
DomTree	1.32 $\times$	1.73 $\times$	2.28 $\times$	0.34 $\times$
KDTree	1.30 $\times$	1.37 $\times$	1.79 $\times$	0.31 $\times$
LinearListReduction	5.74 $\times$	5.69 $\times$	32.70 $\times$	–
List	0.91 $\times$	–	–	–
MonoTree	1.08 $\times$	2.16 $\times$	2.34 $\times$	0.51 $\times$
ObjectGraph	1.07 $\times$	1.98 $\times$	2.12 $\times$	0.65 $\times$
OctTree	–	1.13 $\times$	–	–
PiecewiseFunctions	1.06 $\times$	1.53 $\times$	1.61 $\times$	0.24 $\times$
TernaryTree	1.07 $\times$	1.30 $\times$	1.40 $\times$	0.24 $\times$
Trie	0.96 $\times$	1.42 $\times$	1.37 $\times$	0.27 $\times$

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0123 $\pm$ 0.00001	0.0865 $\pm$ 0.00003	0.0019$\pm$0.00002	<i>OOM</i>	6.41$\times$	7.04$\times$	45.12$\times$	–
branchStatsPass	F	2/9	78%	0.0119 $\pm$ 0.00001	0.1001 $\pm$ 0.00113	0.0034$\pm$0.00000	<i>OOM</i>	3.48$\times$	8.41$\times$	29.30$\times$	–
castInstCountPass	F	3/9	67%	0.0187 $\pm$ 0.00001	0.0994 $\pm$ 0.00115	0.0013$\pm$0.00000	<i>OOM</i>	14.54$\times$	5.31$\times$	77.29$\times$	–
has cycle	F	4/9	56%	0.0254 $\pm$ 0.00079	0.1009 $\pm$ 0.00093	0.0172$\pm$0.00079	<i>OOM</i>	1.48$\times$	3.96$\times$	5.86$\times$	–
instCountPass	F	2/9	78%	0.0123 $\pm$ 0.00042	0.0875 $\pm$ 0.00017	0.0019$\pm$0.00003	<i>OOM</i>	6.40$\times$	7.10$\times$	45.47$\times$	–
latencyModelPass	F	3/9	67%	0.0122 $\pm$ 0.00001	0.0998 $\pm$ 0.00137	0.0030$\pm$0.00000	<i>OOM</i>	4.12$\times$	8.15$\times$	33.56$\times$	–
memoryOpStatsPass	F	3/9	67%	0.0121 $\pm$ 0.00037	0.0993 $\pm$ 0.00113	0.0038$\pm$0.00000	<i>OOM</i>	3.16$\times$	8.22$\times$	25.98$\times$	–
stripSideEffectsPass	M	7/9	22%	0.0804$\pm$0.00141	0.1921 $\pm$ 0.00100	0.0932 $\pm$ 0.00124	<i>OOM</i>	0.86 $\times$	2.39$\times$	2.06$\times$	–
targetReturnPass	M	9/9	0%	0.0790$\pm$0.00016	0.1901 $\pm$ 0.00029	0.0925 $\pm$ 0.00123	<i>OOM</i>	0.85 $\times$	2.41$\times$	2.06$\times$	–
throughputModelPass	F	3/9	67%	0.0123 $\pm$ 0.00000	0.0993 $\pm$ 0.00130	0.0030$\pm$0.00000	<i>OOM</i>	4.13$\times$	8.07$\times$	33.36$\times$	–
verifyIR	F	9/9	0%	0.0127 $\pm$ 0.00015	0.0713 $\pm$ 0.00001	0.0035$\pm$0.00035	<i>OOM</i>	3.66$\times$	5.63$\times$	20.61$\times$	–
Total				0.2894	1.226	0.2247	–	1.29$\times$	4.24$\times$	5.46$\times$	–
Geomean								3.26$\times$	5.55$\times$	18.12$\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	6.2e+07/ 9.7e+06	5.1e+07/ 5.1e+07	1.3e+06/ 1.1e+04	4.2e+02/ 4.1e+01	2.3e+05/ 1.1e+03	4.3e+02/ 4.1e+01	4.6e+06/ 5.7e+02
branchStatsPass	F	6.0e+07/ 1.7e+07	7.9e+07 /8.6e+07	8.4e+05/ 2.4e+04	7.0e+02/ 6.7e+01	5.3e+04/ 3.2e+03	6.9e+02/ 6.9e+01	4.6e+06/ 4.4e+05
castInstCountPass	F	9.5e+07/ 6.5e+06	5.1e+07/ 3.2e+07	1.3e+04 /1.6e+04	6.1e+01/ 3.1e+01	9.2e+02 /1.1e+03	6.2e+01/ 3.2e+01	4.6e+06/ 1.9e+03
has cycle	F	1.3e+08/ 8.7e+07	1.1e+08 /1.5e+08	2.5e+06/ 3.2e+05	7.8e+02/ 1.6e+02	1.8e+05/ 2.2e+04	7.7e+02/ 1.6e+02	5.6e+06/ 2.4e+06
instCountPass	F	6.2e+07/ 9.8e+06	5.6e+07 /5.7e+07	1.3e+06/ 1.0e+04	6.4e+02/ 2.9e+01	2.3e+05/ 1.2e+03	6.4e+02/ 2.8e+01	4.6e+06/ 7.0e+02
latencyModelPass	F	6.2e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.5e+04	4.9e+02/ 7.5e+01	2.1e+05/ 4.0e+03	5.0e+02/ 7.5e+01	4.7e+06/ 5.3e+05
memoryOpStatsPass	F	6.1e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.8e+05/ 3.6e+04	4.8e+02/ 6.8e+01	3.4e+04/ 3.1e+03	4.8e+02/ 6.8e+01	4.6e+06/ 4.4e+05
stripSideEffectsPass	M	1.6e+08 /2.2e+08	1.8e+08 /6.5e+08	1.2e+06 /1.5e+06	1.1e+05 /1.6e+05	6.8e+03 /2.1e+04	5.6e+03 /2.5e+04	4.0e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /2.1e+08	1.7e+08 /7.1e+08	1.7e+06/ 1.5e+06	1.3e+05 /2.1e+05	7.1e+03 /2.5e+04	8.2e+03 /3.1e+04	4.1e+06/ 1.5e+06
throughputModelPass	F	6.2e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.3e+04	4.3e+02/ 5.4e+01	2.1e+05/ 3.5e+03	4.3e+02/ 5.6e+01	4.7e+06/ 5.4e+05
verifyIR	F	6.4e+07/ 1.7e+07	7.4e+07 /8.6e+07	9.6e+05/ 1.9e+04	6.8e+02/ 1.4e+02	1.4e+05/ 4.0e+03	6.8e+02/ 1.4e+02	4.6e+06/ 4.4e+05
Total		9.7e+08/6.3e+08	9.9e+08/2.1e+09	1.3e+07/3.4e+06	2.5e+05/3.7e+05	1.3e+06/9.0e+04	1.9e+04/5.6e+04	5.1e+07/7.4e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.4e+08/–	2.2e+08/–	2.6e+06/–	1.1e+03/–	1.3e+05/–	1.1e+03/–	1.5e+07/–
branchStatsPass	F	5.1e+08/–	2.5e+08/–	2.7e+06/–	6.0e+02/–	8.7e+04/–	6.0e+02/–	1.7e+07/–
castInstCountPass	F	5.0e+08/–	2.5e+08/–	2.7e+06/–	9.2e+02/–	8.9e+04/–	9.2e+02/–	1.7e+07/–
has cycle	F	5.1e+08/–	2.6e+08/–	2.7e+06/–	3.6e+02/–	8.3e+04/–	3.6e+02/–	1.7e+07/–
instCountPass	F	4.4e+08/–	2.2e+08/–	2.6e+06/–	9.2e+02/–	1.3e+05/–	9.1e+02/–	1.4e+07/–
latencyModelPass	F	5.1e+08/–	2.4e+08/–	2.7e+06/–	5.6e+02/–	8.7e+04/–	5.5e+02/–	1.7e+07/–
memoryOpStatsPass	F	5.0e+08/–	2.6e+08/–	2.7e+06/–	7.6e+02/–	8.9e+04/–	7.6e+02/–	1.7e+07/–
stripSideEffectsPass	M	7.1e+08/–	4.2e+08/–	1.6e+06/–	1.3e+05/–	1.7e+05/–	2.6e+04/–	2.4e+07/–
targetReturnPass	M	7.0e+08/–	4.0e+08/–	1.8e+06/–	1.2e+05/–	1.6e+05/–	3.3e+04/–	2.4e+07/–
throughputModelPass	F	5.0e+08/–	2.4e+08/–	2.7e+06/–	6.2e+02/–	8.6e+04/–	6.2e+02/–	1.7e+07/–
verifyIR	F	3.6e+08/–	1.4e+08/–	2.4e+06/–	8.3e+02/–	1.4e+05/–	8.3e+02/–	1.2e+07/–
Total		5.7e+09/–	2.9e+09/–	2.7e+07/–	2.5e+05/–	1.3e+06/–	6.5e+04/–	1.9e+08/–

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0692$\pm$0.00011	0.0786 $\pm$ 0.00009	0.0920 $\pm$ 0.00007	0.3317 $\pm$ 0.00235	0.75 $\times$	1.14$\times$	0.86 $\times$	0.24 $\times$
countJoins	F	3/15	80%	0.0192 $\pm$ 0.00001	0.0296 $\pm$ 0.00001	0.0115$\pm$0.00001	0.1304 $\pm$ 0.00004	1.67$\times$	1.54$\times$	2.57$\times$	0.23 $\times$
filterSelectivitySkew	F	5/15	67%	0.0188 $\pm$ 0.00079	0.0298 $\pm$ 0.00002	0.0124$\pm$0.00001	0.1593 $\pm$ 0.00004	1.51$\times$	1.59$\times$	2.40$\times$	0.19 $\times$
hashJoinPressure	F	5/15	67%	0.0192 $\pm$ 0.00070	0.0298 $\pm$ 0.00001	0.0126$\pm$0.00069	0.1314 $\pm$ 0.00003	1.53$\times$	1.55$\times$	2.37$\times$	0.23 $\times$
scaleCosts	M	15/15	0%	0.0683$\pm$0.00010	0.0786 $\pm$ 0.00107	0.1089 $\pm$ 0.00006	0.3269 $\pm$ 0.00346	0.63 $\times$	1.15$\times$	0.72 $\times$	0.24 $\times$
sumCost	F	6/15	60%	0.0193 $\pm$ 0.00009	0.0291 $\pm$ 0.00020	0.0144$\pm$0.00005	0.1395 $\pm$ 0.00129	1.34$\times$	1.51$\times$	2.02$\times$	0.21 $\times$
sumMemory	F	6/15	60%	0.0190 $\pm$ 0.00001	0.0290 $\pm$ 0.00004	0.0150$\pm$0.00004	0.1389 $\pm$ 0.00172	1.27$\times$	1.52$\times$	1.94$\times$	0.21 $\times$
sumRows	F	6/15	60%	0.0267 $\pm$ 0.00001	0.0298 $\pm$ 0.00002	0.0223$\pm$0.00002	0.1093 $\pm$ 0.00118	1.20$\times$	1.12$\times$	1.34$\times$	0.27 $\times$
Total				0.2598	0.3344	0.2891	1.468	0.90 $\times$	1.29$\times$	1.16$\times$	0.23 $\times$
Geomean								1.18$\times$	1.37$\times$	1.62$\times$	0.22 $\times$

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.7e+08	1.9e+08 /8.9e+08	2.8e+05 /1.3e+06	8.9e+04 /1.8e+05	4.6e+04/ 2.3e+04	4.6e+03 /1.1e+04	2.5e+06/ 8.9e+05
countJoins	F	9.7e+07/ 5.8e+07	7.5e+07 /8.1e+07	6.6e+05/ 3.1e+04	3.6e+02/ 1.2e+02	1.5e+05/ 2.2e+03	3.6e+02/ 6.6e+01	2.6e+06/ 6.4e+02
filterSelectivitySkew	F	9.5e+07/ 6.3e+07	1.0e+08 /1.2e+08	5.3e+05/ 7.5e+04	3.4e+02/ 1.1e+02	1.1e+05/ 6.1e+03	3.4e+02/ 1.1e+02	2.6e+06/ 2.3e+05
hashJoinPressure	F	9.7e+07/ 6.4e+07	8.5e+07 /1.0e+08	8.5e+05/ 7.7e+04	5.1e+02/ 5.5e+01	2.0e+05/ 7.1e+03	5.1e+02/ 5.6e+01	2.7e+06/ 1.2e+05
scaleCosts	M	1.6e+08 /3.5e+08	2.3e+08 /9.2e+08	2.3e+05 /1.5e+06	7.6e+04 /2.9e+05	4.0e+04/ 2.8e+04	5.2e+03 /2.9e+04	2.4e+06/ 1.1e+06
sumCost	F	9.8e+07/ 7.3e+07	9.1e+07 /1.3e+08	5.6e+05/ 7.9e+04	3.0e+02/ 8.4e+01	1.4e+05/ 1.1e+04	3.1e+02/ 8.4e+01	2.7e+06/ 3.3e+05
sumMemory	F	9.6e+07/ 7.6e+07	9.1e+07 /1.3e+08	6.9e+05/ 8.4e+04	4.6e+02/ 5.3e+01	1.6e+05/ 1.1e+04	4.6e+02/ 5.3e+01	2.7e+06/ 3.2e+05
sumRows	F	1.4e+08/ 1.1e+08	2.2e+08 /2.4e+08	8.8e+04 /9.5e+04	2.3e+02/ 8.4e+01	1.4e+04/ 8.9e+03	2.3e+02/ 8.8e+01	2.3e+06/ 2.8e+05
Total		9.5e+08/1.1e+09	1.1e+09/2.6e+09	3.9e+06/3.3e+06	1.7e+05/4.8e+05	8.7e+05/9.8e+04	1.2e+04/4.1e+04	2.0e+07/3.2e+06

Table 11: Per-pass PAPI counters for DBQuery (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.1e+08 /1.5e+09	4.0e+08 /2.7e+09	6.5e+05 /1.9e+07	7.8e+04 /7.0e+05	1.6e+04 /4.6e+04	2.5e+03 /5.5e+04	2.2e+06/ 9.2e+05
countJoins	F	1.5e+08 /6.6e+08	2.4e+08 /9.1e+08	2.0e+06/ 3.0e+05	4.2e+02 /1.8e+03	9.4e+04/ 1.3e+04	4.2e+02/ 1.8e+02	2.3e+06/ 3.0e+04
filterSelectivitySkew	F	1.5e+08 /8.1e+08	2.7e+08 /9.6e+08	1.1e+06/ 9.5e+05	1.5e+02 /2.1e+03	3.8e+04/ 1.7e+04	1.5e+02 /3.0e+02	2.2e+06/ 2.2e+05
hashJoinPressure	F	1.5e+08 /6.7e+08	2.6e+08 /9.1e+08	1.7e+06/ 6.1e+05	1.7e+02 /1.4e+03	5.0e+04/ 1.6e+04	1.7e+02 /2.6e+02	2.2e+06/ 1.1e+05
scaleCosts	M	2.1e+08 /1.5e+09	4.3e+08 /2.7e+09	5.1e+05 /2.3e+07	7.9e+04 /4.9e+05	1.4e+04 /4.7e+04	4.6e+03 /6.2e+04	2.2e+06/ 1.1e+06
sumCost	F	1.5e+08 /7.1e+08	2.6e+08 /9.4e+08	6.4e+05 /1.2e+06	2.2e+02 /1.8e+03	2.0e+04/ 2.0e+04	2.2e+02 /3.9e+02	2.3e+06/ 3.2e+05
sumMemory	F	1.5e+08 /7.1e+08	2.6e+08 /9.4e+08	6.6e+05 /1.2e+06	1.5e+02 /1.9e+03	2.1e+04/ 2.0e+04	1.5e+02 /4.3e+02	2.2e+06/ 3.0e+05
sumRows	F	1.5e+08 /5.5e+08	3.1e+08 /1.0e+09	5.2e+05 /1.3e+06	1.9e+02 /1.6e+03	1.8e+04 /2.1e+04	2.0e+02 /5.4e+02	2.2e+06/ 3.2e+05
Total		1.3e+09/7.0e+09	2.4e+09/1.1e+10	7.8e+06/4.7e+07	1.6e+05/1.2e+06	2.7e+05/2.0e+05	8.4e+03/1.2e+05	1.8e+07/3.3e+06

Table 12: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1 $\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.310 $\pm$ 0.0052	4.412 $\pm$ 0.0057	2.787$\pm$0.0054	24.34 $\pm$ 0.012	1.19$\times$	1.33$\times$	1.58$\times$	0.18 $\times$
classify Depth	F	4/7	43%	0.0473 $\pm$ 0.00078	0.0632 $\pm$ 0.00001	0.0372$\pm$0.00010	0.3473 $\pm$ 0.00163	1.27$\times$	1.33$\times$	1.70$\times$	0.18 $\times$
classify tree	F	5/7	29%	0.0473 $\pm$ 0.00001	0.0631 $\pm$ 0.00001	0.0372$\pm$0.00097	0.3474 $\pm$ 0.00206	1.27$\times$	1.33$\times$	1.70$\times$	0.18 $\times$
countFeatureUses	F	3/7	57%	0.0522 $\pm$ 0.00002	0.1418 $\pm$ 0.00116	0.0324$\pm$0.00003	0.3930 $\pm$ 0.00175	1.61$\times$	2.72$\times$	4.38$\times$	0.36 $\times$
countLeaves	F	2/7	71%	0.0515 $\pm$ 0.00002	0.1435 $\pm$ 0.00160	0.0310$\pm$0.00001	0.3739 $\pm$ 0.00202	1.66$\times$	2.79$\times$	4.64$\times$	0.38 $\times$
countNodes	F	2/7	71%	0.0512 $\pm$ 0.00001	0.1408 $\pm$ 0.00128	0.0311$\pm$0.00003	0.3750 $\pm$ 0.00028	1.64$\times$	2.75$\times$	4.52$\times$	0.38 $\times$
countSmallLeaves	F	3/7	57%	0.0522 $\pm$ 0.00108	0.1411 $\pm$ 0.00118	0.0325$\pm$0.00004	0.3229 $\pm$ 0.00122	1.60$\times$	2.71$\times$	4.34$\times$	0.44 $\times$
inferenceCost	F	2/7	71%	0.0516 $\pm$ 0.00108	0.1434 $\pm$ 0.00003	0.0309$\pm$0.00001	0.3761 $\pm$ 0.00208	1.67$\times$	2.78$\times$	4.64$\times$	0.38 $\times$
sumImpurity	F	3/7	57%	0.0518 $\pm$ 0.00133	0.1398 $\pm$ 0.00087	0.0318$\pm$0.00002	0.3847 $\pm$ 0.00260	1.63$\times$	2.70$\times$	4.40$\times$	0.36 $\times$
sumPathLengths	F	3/7	57%	0.0532 $\pm$ 0.00014	0.1449 $\pm$ 0.00018	0.0338$\pm$0.00001	0.3777 $\pm$ 0.00149	1.57$\times$	2.73$\times$	4.29$\times$	0.38 $\times$
sumSamples	F	3/7	57%	0.0510 $\pm$ 0.00002	0.1391 $\pm$ 0.00003	0.0324$\pm$0.00002	0.3789 $\pm$ 0.00274	1.57$\times$	2.72$\times$	4.29$\times$	0.37 $\times$
treeDepth	F	2/7	71%	0.0521 $\pm$ 0.00001	0.1435 $\pm$ 0.00004	0.0335$\pm$0.00016	0.3761 $\pm$ 0.00367	1.55$\times$	2.76$\times$	4.28$\times$	0.38 $\times$
Total				3.872	5.816	3.151	28.39	1.23$\times$	1.50$\times$	1.85$\times$	0.20 $\times$
Geomean								1.51$\times$	2.29$\times$	3.46$\times$	0.32 $\times$

Table 13: Per-pass PAPI counters for DecisionTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.7e+10/ 1.4e+10	3.6e+10 /5.5e+10	5.1e+08/ 2.8e+07	1.5e+05/ 4.8e+04	1.5e+06 /2.0e+06	1.4e+05/ 4.5e+04	9.2e+08/ 1.8e+07
classify Depth	F	2.4e+08/ 1.9e+08	5.1e+08 /7.8e+08	7.2e+06/ 4.3e+05	1.8e+03/ 6.3e+02	2.1e+04 /2.8e+04	1.8e+03/ 6.1e+02	1.3e+07/ 2.8e+05
classify tree	F	2.4e+08/ 1.9e+08	5.1e+08 /7.8e+08	7.3e+06/ 4.3e+05	2.0e+03/ 5.4e+02	2.3e+04 /2.8e+04	2.0e+03/ 5.2e+02	1.3e+07/ 2.8e+05
countFeatureUses	F	2.6e+08/ 1.6e+08	7.7e+08 /8.4e+08	6.5e+06/ 4.0e+05	2.7e+03/ 4.8e+02	3.7e+04/ 1.2e+04	2.7e+03/ 4.9e+02	1.5e+07/ 2.0e+06
countLeaves	F	2.6e+08/ 1.6e+08	5.7e+08/ 5.5e+08	6.4e+06/ 1.2e+05	3.1e+03/ 8.0e+02	4.4e+04/ 8.5e+03	3.0e+03/ 7.7e+02	1.4e+07/ 3.4e+05
countNodes	F	2.6e+08/ 1.6e+08	5.7e+08/ 5.5e+08	6.6e+06/ 1.2e+05	3.1e+03/ 8.4e+02	4.6e+04/ 8.5e+03	3.1e+03/ 8.0e+02	1.4e+07/ 3.5e+05
countSmallLeaves	F	2.6e+08/ 1.6e+08	6.6e+08 /7.8e+08	5.4e+06/ 8.5e+04	2.7e+03/ 5.9e+02	3.8e+04/ 1.3e+04	2.7e+03/ 5.8e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.6e+08/ 1.6e+08	6.9e+08/ 6.9e+08	8.9e+06/ 1.4e+05	2.2e+03/ 6.1e+02	2.1e+04/ 8.1e+03	2.2e+03/ 6.0e+02	1.4e+07/ 3.3e+05
sumImpurity	F	2.6e+08/ 1.6e+08	7.0e+08 /7.8e+08	6.8e+06/ 3.3e+05	2.4e+03/ 6.3e+02	3.7e+04/ 1.2e+04	2.4e+03/ 6.3e+02	1.4e+07/ 2.0e+06
sumPathLengths	F	2.7e+08/ 1.7e+08	7.4e+08 /8.7e+08	7.8e+06/ 7.9e+05	3.0e+03/ 6.0e+02	4.8e+04/ 1.3e+04	3.0e+03/ 6.1e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.6e+08/ 1.6e+08	5.9e+08 /7.1e+08	6.1e+06/ 2.1e+05	2.6e+03/ 1.6e+03	4.2e+04/ 1.5e+04	2.5e+03/ 1.6e+03	1.5e+07/ 2.0e+06
treeDepth	F	2.6e+08/ 1.7e+08	6.9e+08/ 6.9e+08	8.6e+06/ 1.4e+05	1.6e+03/ 4.6e+02	1.8e+04/ 7.8e+03	1.6e+03/ 4.7e+02	1.4e+07/ 3.3e+05
Total		2.0e+10/1.6e+10	4.3e+10/6.3e+10	5.9e+08/3.1e+07	1.7e+05/5.5e+04	1.9e+06/2.1e+06	1.7e+05/5.3e+04	1.1e+09/3.0e+07

Table 14: Per-pass PAPI counters for DecisionTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.2e+10 /1.2e+11	6.6e+10 /2.4e+11	5.1e+07/ 4.8e+07	3.1e+04 /3.4e+04	3.4e+05 /2.2e+06	3.0e+04/ 2.7e+04	8.4e+08/ 1.8e+07
classify Depth	F	3.2e+08 /1.8e+09	9.4e+08 /3.4e+09	4.6e+05 /6.8e+05	3.8e+02/ 3.7e+02	7.0e+03 /3.2e+04	3.8e+02/ 3.3e+02	1.2e+07/ 3.0e+05
classify tree	F	3.2e+08 /1.8e+09	9.4e+08 /3.4e+09	4.6e+05 /6.8e+05	3.4e+02 /3.7e+02	6.9e+03 /3.2e+04	3.4e+02/ 3.2e+02	1.2e+07/ 3.0e+05
countFeatureUses	F	7.2e+08 /2.0e+09	1.9e+09 /4.0e+09	2.1e+06 /3.7e+06	5.4e+02 /1.2e+03	4.9e+03 /7.5e+04	5.4e+02/ 4.2e+02	9.2e+06/ 1.2e+06
countLeaves	F	7.3e+08 /1.9e+09	1.7e+09 /3.8e+09	6.2e+06/ 7.3e+05	5.0e+02 /2.1e+03	8.0e+03 /3.4e+04	4.9e+02/ 3.6e+02	9.1e+06/ 3.1e+05
countNodes	F	7.1e+08 /1.9e+09	1.7e+09 /3.8e+09	3.3e+06/ 7.4e+05	9.2e+02 /1.5e+03	7.5e+03 /3.4e+04	9.0e+02/ 2.0e+02	9.3e+06/ 3.2e+05
countSmallLeaves	F	7.2e+08 /1.6e+09	1.8e+09 /3.9e+09	1.2e+06 /3.6e+06	3.1e+02 /2.1e+03	4.4e+03 /7.8e+04	3.2e+02 /1.0e+03	9.3e+06/ 1.2e+06
inferenceCost	F	7.3e+08 /1.9e+09	1.8e+09 /3.8e+09	5.3e+06/ 7.2e+05	5.0e+02 /2.6e+03	7.9e+03 /3.4e+04	5.1e+02/ 2.4e+02	9.1e+06/ 3.1e+05
sumImpurity	F	7.1e+08 /2.0e+09	1.8e+09 /3.8e+09	1.0e+06 /3.6e+06	1.9e+02 /4.8e+02	4.1e+03 /7.3e+04	1.9e+02 /4.3e+02	9.3e+06/ 1.1e+06
sumPathLengths	F	7.4e+08 /1.9e+09	1.9e+09 /3.9e+09	1.5e+06 /3.7e+06	7.3e+02/ 3.7e+02	4.7e+03 /7.9e+04	7.3e+02/ 3.5e+02	9.1e+06/ 1.1e+06
sumSamples	F	7.0e+08 /1.9e+09	1.7e+09 /3.8e+09	9.3e+05 /3.7e+06	7.4e+02 /1.1e+03	4.7e+03 /7.9e+04	7.4e+02/ 4.5e+02	9.3e+06/ 1.2e+06
treeDepth	F	7.3e+08 /1.9e+09	1.8e+09 /3.8e+09	5.3e+06/ 7.2e+05	6.5e+02 /2.3e+03	7.4e+03 /3.4e+04	6.6e+02/ 3.7e+02	9.1e+06/ 3.2e+05
Total		2.9e+10/1.4e+11	8.4e+10/2.8e+11	7.8e+07/7.1e+07	3.7e+04/4.8e+04	4.1e+05/2.8e+06	3.6e+04/3.2e+04	9.4e+08/2.6e+07

Table 15: Per-pass performance for DomTree, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1 $\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0359 $\pm$ 0.00070	0.0715 $\pm$ 0.00083	0.0189$\pm$0.00005	0.1969 $\pm$ 0.00130	1.91$\times$	1.99$\times$	3.79$\times$	0.36 $\times$
computeWidths	M	14/15	7%	0.0298$\pm$0.00004	0.0314 $\pm$ 0.00004	0.0487 $\pm$ 0.00106	0.0991 $\pm$ 0.00094	0.61 $\times$	1.06 $\times$	0.65 $\times$	0.32 $\times$
count styled	F	3/15	80%	0.0350 $\pm$ 0.00001	0.0705 $\pm$ 0.00001	0.0117$\pm$0.00002	0.1850 $\pm$ 0.00129	3.00$\times$	2.02$\times$	6.05$\times$	0.38 $\times$
find max Bottom	F	5/15	67%	0.0348 $\pm$ 0.00001	0.0700 $\pm$ 0.00202	0.0202$\pm$0.00001	0.1997 $\pm$ 0.00119	1.73$\times$	2.01$\times$	3.47$\times$	0.35 $\times$
scaleLayout	M	15/15	0%	0.0287$\pm$0.00004	0.0306 $\pm$ 0.00004	0.0398 $\pm$ 0.00004	0.0898 $\pm$ 0.00116	0.72 $\times$	1.07 $\times$	0.77 $\times$	0.34 $\times$
sumTextWidth	F	3/15	80%	0.0346 $\pm$ 0.00001	0.0696 $\pm$ 0.00113	0.0116$\pm$0.00001	0.2551 $\pm$ 0.00019	2.99$\times$	2.01$\times$	6.00$\times$	0.27 $\times$
Total				0.1988	0.3437	0.1508	1.026	1.32$\times$	1.73$\times$	2.28$\times$	0.34 $\times$
Geomean								1.53$\times$	1.62$\times$	2.49$\times$	0.34 $\times$

Table 16: Per-pass PAPI counters for DomTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.8e+08/ 9.6e+07	2.7e+08 /4.2e+08	2.4e+06/ 8.1e+05	2.3e+03/ 6.9e+01	1.8e+05/ 1.2e+04	2.3e+03/ 6.9e+01	1.4e+07/ 1.9e+06
computeWidths	M	6.6e+07 /1.6e+08	1.4e+08 /5.8e+08	1.3e+06 /1.4e+06	4.1e+04 /6.6e+04	4.4e+03 /1.3e+04	3.1e+03 /1.6e+04	1.1e+06/ 4.2e+05
count styled	F	1.8e+08/ 5.9e+07	2.5e+08 /2.6e+08	2.2e+06/ 2.7e+04	2.9e+03/ 6.4e+01	2.8e+05/ 4.9e+03	2.8e+03/ 6.4e+01	1.4e+07/ 6.4e+05
find max Bottom	F	1.8e+08/ 1.0e+08	2.6e+08 /4.7e+08	2.3e+06/ 1.3e+06	5.1e+02/ 9.8e+01	1.7e+05/ 9.6e+03	5.1e+02/ 9.7e+01	1.4e+07/ 1.3e+06
scaleLayout	M	6.0e+07 /1.2e+08	8.8e+07 /3.6e+08	1.0e+06 /1.3e+06	7.8e+04/ 5.3e+04	1.5e+04/ 1.2e+04	4.4e+03 /1.3e+04	1.2e+06/ 4.4e+05
sumTextWidth	F	1.8e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.5e+06/ 2.1e+04	2.2e+03/ 9.3e+01	4.6e+05/ 4.3e+03	2.2e+03/ 9.1e+01	1.3e+07/ 6.5e+05
Total		8.4e+08/5.9e+08	1.2e+09/2.3e+09	1.2e+07/4.8e+06	1.3e+05/1.2e+05	1.1e+06/5.6e+04	1.5e+04/2.9e+04	5.7e+07/5.3e+06

Table 17: Per-pass PAPI counters for DomTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.6e+08 /1.0e+09	6.4e+08 /2.1e+09	8.5e+06/ 3.3e+06	4.2e+02 /1.7e+03	1.2e+04 /5.4e+04	4.2e+02 /1.7e+03	9.9e+06/ 1.3e+06
computeWidths	M	7.3e+07 /4.1e+08	1.8e+08 /1.0e+09	1.3e+06 /3.4e+06	5.6e+04 /8.4e+04	4.5e+03 /2.4e+04	2.2e+03 /2.2e+04	1.1e+06/ 4.4e+05
count styled	F	3.6e+08 /9.4e+08	6.3e+08 /2.1e+09	9.9e+06/ 1.3e+06	2.9e+02 /1.7e+03	1.5e+04 /3.9e+04	2.9e+02/ 2.8e+02	1.0e+07/ 4.1e+05
find max Bottom	F	3.6e+08 /1.0e+09	6.6e+08 /2.2e+09	8.9e+06/ 3.0e+06	4.6e+02 /1.8e+03	1.2e+04 /5.7e+04	4.6e+02 /1.4e+03	1.0e+07/ 1.0e+06
scaleLayout	M	6.9e+07 /3.7e+08	1.3e+08 /8.5e+08	9.1e+05 /2.9e+06	4.2e+04 /1.1e+05	3.7e+03 /1.6e+04	2.1e+03 /2.2e+04	1.1e+06/ 4.6e+05
sumTextWidth	F	3.5e+08 /1.3e+09	5.8e+08 /2.1e+09	8.6e+06/ 1.3e+06	2.3e+02 /2.3e+03	5.6e+04/ 4.0e+04	2.3e+02 /2.8e+02	1.0e+07/ 4.1e+05
Total		1.6e+09/5.0e+09	2.8e+09/1.0e+10	3.8e+07/1.5e+07	9.9e+04/2.0e+05	1.0e+05/2.3e+05	5.7e+03/4.7e+04	4.2e+07/4.0e+06

Table 18: Per-pass performance for KDTree, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0278 $\pm$ 0.00003	0.0409 $\pm$ 0.00005	0.0212$\pm$0.00002	0.1334 $\pm$ 0.00109	1.31$\times$	1.47$\times$	1.93$\times$	0.31 $\times$
countInRange tight_box	F	13/17	24%	0.0285 $\pm$ 0.00001	0.0411 $\pm$ 0.00002	0.0208$\pm$0.00001	0.1320 $\pm$ 0.00001	1.37$\times$	1.44$\times$	1.97$\times$	0.31 $\times$
photonMappingBenchmark	F	12/17	29%	0.0440 $\pm$ 0.00001	0.0491 $\pm$ 0.00002	0.0409$\pm$0.00001	0.1414 $\pm$ 0.00020	1.08 $\times$	1.12$\times$	1.20$\times$	0.35 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0252 $\pm$ 0.00074	0.0386 $\pm$ 0.00001	0.0162$\pm$0.00000	0.1299 $\pm$ 0.00114	1.55$\times$	1.53$\times$	2.38$\times$	0.30 $\times$
sumMassInRange	F	14/17	18%	0.0284 $\pm$ 0.00001	0.0413 $\pm$ 0.00001	0.0218$\pm$0.00001	0.1334 $\pm$ 0.00084	1.30$\times$	1.46$\times$	1.90$\times$	0.31 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0294 $\pm$ 0.00001	0.0409 $\pm$ 0.00097	0.0201$\pm$0.00013	0.1316 $\pm$ 0.00001	1.47$\times$	1.39$\times$	2.04$\times$	0.31 $\times$
Total				0.1833	0.2519	0.1410	0.8018	1.30$\times$	1.37$\times$	1.79$\times$	0.31 $\times$
Geomean								1.34$\times$	1.39$\times$	1.87$\times$	0.31 $\times$

Table 19: Per-pass PAPI counters for KDTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	5.1e+08 /6.4e+08	3.3e+06/ 2.3e+06	1.0e+02/ 6.9e+01	5.3e+03 /1.9e+04	1.0e+02/ 6.9e+01	7.7e+06/ 1.6e+06
countInRange tight_box	F	1.4e+08/ 1.1e+08	4.6e+08 /6.0e+08	3.1e+06/ 2.4e+06	7.0e+01/ 6.8e+01	5.0e+03 /2.0e+04	7.4e+01/ 6.8e+01	7.7e+06/ 1.7e+06
photonMappingBenchmark	F	2.2e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.0e+06 /2.6e+06	4.8e+02/ 4.0e+02	2.9e+03 /1.6e+04	4.8e+02/ 4.0e+02	5.4e+06/ 1.4e+06
pointCloudNeighborhood	F	1.3e+08/ 8.2e+07	3.5e+08 /4.7e+08	2.9e+06/ 2.1e+06	1.4e+02 /1.6e+02	5.1e+03 /1.8e+04	1.5e+02 /1.6e+02	7.7e+06/ 1.4e+06
sumMassInRange	F	1.4e+08/ 1.1e+08	4.6e+08 /6.2e+08	3.1e+06 /3.6e+06	1.6e+02/ 1.5e+02	4.8e+03 /2.6e+04	1.6e+02/ 1.5e+02	7.6e+06/ 1.8e+06
twoPointCorrelation bin_8_16	F	1.5e+08/ 1.0e+08	5.1e+08 /6.1e+08	1.9e+06/ 7.9e+05	6.5e+01 /1.7e+02	1.9e+04/ 1.2e+04	6.5e+01 /1.7e+02	7.6e+06/ 1.3e+06
Total		9.3e+08/7.2e+08	3.3e+09/4.1e+09	1.6e+07/1.4e+07	1.0e+03/1.0e+03	4.2e+04/1.1e+05	1.0e+03/1.0e+03	4.4e+07/9.3e+06

Table 20: Per-pass PAPI counters for KDTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.1e+08 /6.8e+08	6.2e+08 /1.5e+09	4.8e+06 /5.9e+06	2.6e+02 /2.0e+03	5.4e+03 /2.3e+04	2.6e+02 /1.9e+03	5.7e+06/ 1.5e+06
countInRange tight_box	F	2.1e+08 /6.7e+08	5.4e+08 /1.3e+09	5.0e+06 /5.8e+06	1.2e+02 /1.9e+03	4.8e+03 /2.0e+04	1.2e+02 /1.9e+03	5.7e+06/ 1.5e+06
photonMappingBenchmark	F	2.5e+08 /7.2e+08	1.2e+09 /2.1e+09	2.3e+06 /5.6e+06	3.8e+02 /2.3e+03	3.1e+03 /2.3e+04	3.8e+02 /2.3e+03	5.1e+06/ 1.6e+06
pointCloudNeighborhood	F	2.0e+08 /6.6e+08	5.0e+08 /1.4e+09	4.3e+06 /4.9e+06	8.6e+01 /1.6e+03	5.7e+03 /1.8e+04	8.8e+01 /1.6e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.1e+08 /6.8e+08	5.6e+08 /1.4e+09	5.1e+06 /6.6e+06	7.7e+01 /1.5e+03	4.8e+03 /2.1e+04	7.7e+01 /1.5e+03	5.6e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.1e+08 /6.7e+08	6.8e+08 /1.5e+09	5.1e+06/ 3.9e+06	3.0e+02 /1.5e+03	9.3e+03 /1.7e+04	3.0e+02 /1.5e+03	5.7e+06/ 1.2e+06
Total		1.3e+09/4.1e+09	4.1e+09/9.2e+09	2.7e+07/3.3e+07	1.2e+03/1.1e+04	3.3e+04/1.2e+05	1.2e+03/1.1e+04	3.4e+07/8.8e+06

Table 21: Per-pass performance for LinearListReduction, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0286 $\pm$ 0.00002	0.1628 $\pm$ 0.00125	0.0050$\pm$0.00002	<i>OOM</i>	5.74$\times$	5.69$\times$	32.70$\times$	–
Total				0.0286	0.1628	0.0050	–	5.74$\times$	5.69$\times$	32.70$\times$	–
Geomean								5.74$\times$	5.69$\times$	32.70$\times$	–

Table 22: Per-pass PAPI counters for LinearListReduction (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	4.7e+06/ 5.8e+04	7.6e+02/ 4.2e+01	1.9e+06/ 7.3e+03	7.6e+02/ 4.3e+01	1.2e+07/ 1.1e+06
Total		1.5e+08/2.5e+07	9.0e+07/1.0e+08	4.7e+06/5.8e+04	7.6e+02/4.2e+01	1.9e+06/7.3e+03	7.6e+02/4.3e+01	1.2e+07/1.1e+06

Table 23: Per-pass PAPI counters for LinearListReduction (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	8.3e+08/–	3.7e+08/–	5.0e+06/–	2.8e+03/–	3.3e+05/–	2.8e+03/–	3.0e+07/–
Total		8.3e+08/–	3.7e+08/–	5.0e+06/–	2.8e+03/–	3.3e+05/–	2.8e+03/–	3.0e+07/–

Table 24: Per-pass performance for `List`, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2548±0.00158	<i>OOM</i>	0.3122±0.00115	<i>OOM</i>	0.82×	–	–	–
length List	M	1/2	50%	0.0413±0.00002	<i>OOM</i>	0.0216±0.00001	<i>OOM</i>	1.91×	–	–	–
sumList	F	2/2	0%	0.0440±0.00001	<i>OOM</i>	0.0453±0.00001	<i>OOM</i>	0.97×	–	–	–
sumList tail recursive	F	2/2	0%	0.0440±0.00001	<i>OOM</i>	0.0450±0.00001	<i>OOM</i>	0.98×	–	–	–
Total				0.3840	–	0.4241	–	0.91×	–	–	–
Geomean								1.10×	–	–	–

Table 25: Per-pass PAPI counters for `List` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	6.0e+08 /8.3e+08	1.9e+09 /3.1e+09	7.8e+06/ 1.1e+06	3.1e+05 /5.7e+05	1.2e+04 /5.5e+04	1.4e+04/ 1.4e+04	9.1e+06/ 8.1e+06
length List	M	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.3e+05 /2.8e+05	1.9e+02/ 7.4e+01	3.4e+03 /1.8e+04	1.9e+02/ 7.4e+01	1.3e+07/ 1.2e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.1e+09	9.5e+05/ 6.2e+05	2.5e+02/ 7.8e+01	4.1e+03 /5.9e+04	2.5e+02/ 7.9e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.1e+09	9.5e+05/ 7.0e+05	1.8e+02/ 1.3e+02	4.1e+03 /5.3e+04	1.9e+02/ 1.3e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.4e+09	4.5e+09/6.1e+09	9.8e+06/2.7e+06	3.1e+05/5.7e+05	2.4e+04/1.8e+05	1.5e+04/1.4e+04	4.9e+07/3.4e+07

Table 26: Per-pass PAPI counters for `List` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 27: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0463±0.00130	0.0658±0.00117	0.0421±0.00041	0.1299±0.00010	1.10×	1.42×	1.56×	0.51×
sumTree	F	3/3	0%	0.0116±0.00001	0.0424±0.00001	0.0113±0.00001	0.0813±0.00172	1.02×	3.67×	3.74×	0.52×
sumTree TailRec	F	3/3	0%	0.0120±0.00000	0.0425±0.00001	0.0111±0.00001	0.0820±0.00005	1.08×	3.54×	3.82×	0.52×
Total				0.0699	0.1507	0.0645	0.2933	1.08×	2.16×	2.34×	0.51×
Geomean								1.07×	2.64×	2.82×	0.52×

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.7e+08/ 1.4e+08	5.0e+08 /6.4e+08	1.5e+05 /4.4e+05	4.6e+04 /7.4e+04	2.4e+03 /5.8e+03	3.7e+03 /3.9e+03	5.0e+05/ 4.6e+05
sumTree	F	5.8e+07/ 5.8e+07	2.1e+08 /2.5e+08	3.9e+03 /4.9e+04	2.0e+02 /5.2e+02	4.7e+02 /4.3e+03	2.0e+02 /5.2e+02	7.1e+05/ 6.2e+05
sumTree TailRec	F	6.1e+07/ 5.6e+07	1.9e+08 /2.2e+08	3.9e+03 /3.4e+04	2.2e+02 /5.1e+02	5.3e+02 /4.1e+03	2.2e+02 /5.1e+02	6.9e+05/ 6.1e+05
Total		2.9e+08/2.6e+08	9.1e+08/1.1e+09	1.6e+05/5.3e+05	4.7e+04/7.6e+04	3.4e+03/1.4e+04	4.1e+03/4.9e+03	1.9e+06/1.7e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.7e+08 /5.9e+08	9.1e+08 /1.4e+09	2.4e+05 /1.1e+06	3.6e+04 /6.6e+04	1.9e+03 /2.2e+04	3.0e+03/ 2.5e+03	4.7e+05/ 3.7e+05
sumTree	F	2.2e+08 /4.1e+08	6.0e+08 /7.6e+08	9.2e+04 /1.1e+06	2.8e+02 /3.6e+02	8.3e+02 /1.8e+04	2.8e+02/ 2.7e+02	3.7e+05/ 3.5e+05
sumTree TailRec	F	2.2e+08 /4.2e+08	6.0e+08 /7.6e+08	9.8e+04 /1.1e+06	1.7e+02/ 9.5e+01	8.8e+02 /1.8e+04	1.7e+02/ 9.6e+01	3.5e+05 /3.6e+05
Total		7.0e+08/1.4e+09	2.1e+09/2.9e+09	4.3e+05/3.4e+06	3.7e+04/6.7e+04	3.6e+03/5.8e+04	3.5e+03/2.9e+03	1.2e+06/1.1e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0145±0.00000	0.0452±0.00001	0.0109±0.00001	0.0523±0.00125	1.33×	3.11×	4.14×	0.86×
countMarked	F	3/5	40%	0.0147±0.00000	0.0455±0.00001	0.0108±0.00002	0.0502±0.00109	1.36×	3.09×	4.20×	0.91×
countSurvivors	F	4/5	20%	0.0147±0.00000	0.0475±0.00001	0.0128±0.00000	0.0520±0.00102	1.15×	3.23×	3.72×	0.91×
deadBytes	F	4/5	20%	0.0154±0.00000	0.0464±0.00090	0.0127±0.00001	0.0512±0.00001	1.21×	3.02×	3.66×	0.91×
liveBytes	F	4/5	20%	0.0155±0.00069	0.0464±0.00000	0.0127±0.00001	0.0512±0.00001	1.22×	2.99×	3.65×	0.91×
sumObjIds	F	3/5	40%	0.0142±0.00001	0.0441±0.00002	0.0102±0.00002	0.0506±0.00001	1.39×	3.10×	4.31×	0.87×
sweepUnmarked	M	5/5	0%	0.0738±0.00006	0.0919±0.00007	0.0791±0.00105	0.2025±0.00022	0.93×	1.25×	1.16×	0.45×
totalHeapSize	F	3/5	40%	0.0142±0.00017	0.0440±0.00016	0.0106±0.00101	0.0508±0.00003	1.35×	3.09×	4.16×	0.87×
touchHotObjects	M	5/5	0%	0.0767±0.00006	0.0913±0.00022	0.0774±0.00004	0.2119±0.00155	0.99×	1.19×	1.18×	0.43×
Total				0.2538	0.5022	0.2371	0.7727	1.07×	1.98×	2.12×	0.65×
Geomean								1.20×	2.51×	3.03×	0.76×

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	7.4e+07/ 5.5e+07	2.7e+08 /3.0e+08	1.9e+05/ 4.2e+04	4.6e+02/ 3.5e+02	3.0e+03 /3.8e+03	4.6e+02/ 3.5e+02	2.6e+06/ 6.4e+05
countMarked	F	7.5e+07/ 5.5e+07	2.7e+08 /3.0e+08	1.9e+05/ 4.2e+04	4.4e+02/ 4.1e+02	2.7e+03 /4.4e+03	4.3e+02/ 4.1e+02	2.6e+06/ 6.6e+05
countSurvivors	F	7.5e+07/ 6.5e+07	3.0e+08 /3.5e+08	1.7e+05/ 7.2e+04	4.4e+02/ 3.0e+02	2.6e+03 /5.6e+03	4.4e+02/ 3.0e+02	2.6e+06/ 1.2e+06
deadBytes	F	7.8e+07/ 6.4e+07	2.6e+08 /3.3e+08	2.4e+05/ 8.2e+04	4.5e+02/ 4.0e+02	2.5e+03 /6.1e+03	4.5e+02/ 4.0e+02	2.6e+06/ 1.2e+06
liveBytes	F	7.8e+07/ 6.4e+07	2.6e+08 /3.3e+08	2.2e+05/ 8.3e+04	2.1e+02 /2.5e+02	1.9e+03 /6.0e+03	2.2e+02 /2.5e+02	2.6e+06/ 1.2e+06
sumObjIds	F	7.2e+07/ 5.2e+07	2.5e+08 /2.7e+08	3.0e+05/ 2.9e+04	7.0e+02/ 4.4e+02	3.6e+03 /4.8e+03	7.0e+02/ 4.4e+02	2.6e+06/ 6.4e+05
sweepUnmarked	M	2.0e+08 /2.3e+08	5.5e+08 /9.4e+08	8.6e+05/ 4.4e+05	9.3e+04 /1.2e+05	6.0e+03 /1.2e+04	1.1e+04/ 9.2e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	7.2e+07/ 5.3e+07	2.5e+08 /2.7e+08	3.0e+05/ 3.5e+04	5.5e+02/ 4.6e+02	3.2e+03 /3.9e+03	5.5e+02/ 4.5e+02	2.6e+06/ 6.5e+05
touchHotObjects	M	2.2e+08 /2.2e+08	5.5e+08 /9.3e+08	5.5e+05/ 5.2e+05	7.8e+04 /1.1e+05	5.3e+03 /1.1e+04	8.1e+03/ 6.1e+03	1.9e+06/ 1.0e+06
Total		9.4e+08/8.6e+08	3.0e+09/4.0e+09	3.0e+06/1.3e+06	1.7e+05/2.4e+05	3.1e+04/5.7e+04	2.2e+04/1.8e+04	2.2e+07/8.3e+06

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.7e+08	6.5e+08 /1.2e+09	9.2e+04 /1.1e+06	6.7e+01 /3.0e+02	1.0e+03 /1.7e+04	6.8e+01 /3.0e+02	1.8e+06/ 3.5e+05
countMarked	F	2.3e+08 /2.5e+08	6.5e+08 /1.1e+09	1.6e+05 /1.0e+06	1.3e+03/ 2.0e+02	2.5e+03 /1.6e+04	1.3e+03/ 2.1e+02	1.8e+06/ 3.6e+05
countSurvivors	F	2.4e+08 /2.6e+08	6.8e+08 /1.2e+09	1.2e+05 /1.7e+06	2.4e+02 /2.7e+02	1.3e+03 /7.1e+03	2.5e+02 /2.7e+02	1.7e+06/ 5.8e+05
deadBytes	F	2.4e+08 /2.6e+08	6.6e+08 /1.1e+09	1.3e+05 /1.3e+06	1.4e+02 /2.2e+02	1.3e+03 /8.4e+03	1.5e+02 /2.2e+02	1.8e+06/ 6.0e+05
liveBytes	F	2.4e+08 /2.6e+08	6.6e+08 /1.1e+09	1.3e+05 /1.3e+06	2.2e+02/ 2.1e+02	1.2e+03 /8.4e+03	2.2e+02/ 2.1e+02	1.8e+06/ 6.1e+05
sumObjIds	F	2.2e+08 /2.6e+08	6.4e+08 /1.1e+09	9.7e+04 /1.0e+06	9.0e+01 /2.1e+02	1.0e+03 /1.4e+04	9.2e+01 /2.1e+02	1.7e+06/ 3.5e+05
sweepUnmarked	M	3.0e+08 /8.5e+08	9.6e+08 /2.3e+09	2.1e+05 /2.5e+06	8.1e+04 /1.5e+05	4.4e+03 /2.3e+04	4.5e+03 /7.1e+03	1.8e+06/ 9.4e+05
totalHeapSize	F	2.2e+08 /2.6e+08	6.4e+08 /1.1e+09	9.4e+04 /1.0e+06	6.7e+01 /2.8e+02	9.7e+02 /1.7e+04	6.8e+01 /2.8e+02	1.8e+06/ 3.6e+05
touchHotObjects	M	2.9e+08 /9.0e+08	9.7e+08 /2.4e+09	1.7e+05 /2.5e+06	9.2e+04 /1.7e+05	4.1e+03 /2.3e+04	4.3e+03 /8.3e+03	1.8e+06/ 9.3e+05
Total		2.2e+09/3.6e+09	6.5e+09/1.3e+10	1.2e+06/1.3e+07	1.8e+05/3.3e+05	1.8e+04/1.3e+05	1.1e+04/1.7e+04	1.6e+07/5.1e+06

Table 33: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1195$\pm$0.00106	0.1351 $\pm$ 0.00013	0.1346 $\pm$ 0.00107	0.4465 $\pm$ 0.00163	0.89 $\times$	1.13$\times$	1.00 $\times$	0.30 $\times$
autorefineMaxLevel	F	4/8	50%	0.0218 $\pm$ 0.00072	0.0496 $\pm$ 0.00001	0.0200$\pm$0.00012	0.2724 $\pm$ 0.00157	1.09 $\times$	2.27$\times$	2.48$\times$	0.18 $\times$
compressMass	F	3/8	62%	0.0198 $\pm$ 0.00001	0.0468 $\pm$ 0.00001	0.0116$\pm$0.00001	0.2352 $\pm$ 0.00118	1.70$\times$	2.37$\times$	4.03$\times$	0.20 $\times$
diffPW	M	8/8	0%	0.1201$\pm$0.00015	0.1367 $\pm$ 0.00127	0.1343 $\pm$ 0.00116	0.4474 $\pm$ 0.00120	0.89 $\times$	1.14$\times$	1.02 $\times$	0.31 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0214 $\pm$ 0.00000	0.0498 $\pm$ 0.00001	0.0169$\pm$0.00002	0.2636 $\pm$ 0.00117	1.27$\times$	2.32$\times$	2.95$\times$	0.19 $\times$
norm2Estimate	F	5/8	38%	0.0255 $\pm$ 0.00016	0.0578 $\pm$ 0.00005	0.0181$\pm$0.00005	0.2596 $\pm$ 0.00226	1.41$\times$	2.26$\times$	3.19$\times$	0.22 $\times$
pmapCutHistogram	F	4/8	50%	0.0321 $\pm$ 0.00001	0.0577 $\pm$ 0.00117	0.0126$\pm$0.00000	0.2672 $\pm$ 0.00013	2.55$\times$	1.80$\times$	4.58$\times$	0.22 $\times$
truncateTolViolations	F	3/8	62%	0.0201 $\pm$ 0.00000	0.0475 $\pm$ 0.00101	0.0117$\pm$0.00001	0.2267 $\pm$ 0.00132	1.73$\times$	2.36$\times$	4.07$\times$	0.21 $\times$
Total				0.3803	0.5810	0.3598	2.419	1.06 $\times$	1.53$\times$	1.61$\times$	0.24 $\times$
Geomean								1.36$\times$	1.88$\times$	2.55$\times$	0.22 $\times$

Table 34: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.8e+08 /3.6e+08	5.9e+08 /1.4e+09	1.2e+06 /2.2e+06	3.1e+05/ 2.1e+05	1.3e+04 /2.8e+04	1.8e+04 /3.0e+04	4.4e+06/ 1.8e+06
autorefineMaxLevel	F	1.1e+08/ 1.0e+08	3.1e+08 /4.6e+08	3.1e+06/ 2.5e+05	1.0e+03/ 5.8e+01	5.9e+03 /6.3e+03	1.0e+03/ 5.9e+01	6.0e+06/ 9.7e+05
compressMass	F	1.0e+08/ 5.9e+07	2.4e+08 /2.8e+08	2.1e+06/ 3.3e+04	1.2e+03/ 2.4e+02	1.8e+04/ 4.6e+03	1.2e+03/ 2.4e+02	6.2e+06/ 6.1e+05
diffPW	M	2.8e+08 /3.6e+08	6.0e+08 /1.4e+09	7.3e+05 /1.9e+06	1.7e+05 /2.1e+05	1.4e+04 /2.5e+04	1.5e+04 /3.1e+04	4.5e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 8.6e+07	3.3e+08 /4.5e+08	2.2e+06/ 3.2e+05	1.1e+03/ 1.5e+02	1.7e+04/ 8.7e+03	1.1e+03/ 1.5e+02	6.2e+06/ 1.7e+06
norm2Estimate	F	1.3e+08/ 9.2e+07	3.5e+08 /4.4e+08	1.6e+05 /5.1e+05	9.9e+02/ 1.8e+02	6.1e+03 /9.4e+03	9.9e+02/ 1.8e+02	5.4e+06/ 1.5e+06
pmapCutHistogram	F	1.6e+08/ 6.4e+07	2.7e+08 /3.3e+08	9.3e+05/ 4.9e+04	1.7e+03/ 2.8e+02	5.6e+04/ 7.0e+03	1.7e+03/ 2.8e+02	4.6e+06/ 1.3e+06
truncateTolViolations	F	1.0e+08/ 5.9e+07	2.4e+08 /2.8e+08	2.1e+06/ 4.9e+04	1.0e+03/ 2.9e+02	1.6e+04/ 4.9e+03	1.0e+03/ 2.9e+02	6.3e+06/ 6.1e+05
Total		1.3e+09/1.2e+09	2.9e+09/5.0e+09	1.2e+07/5.3e+06	4.9e+05/4.2e+05	1.5e+05/9.3e+04	4.0e+04/6.2e+04	4.4e+07/1.0e+07

Table 35: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.5e+08 /1.9e+09	9.8e+08 /3.5e+09	4.0e+05 /5.0e+06	1.6e+05 /2.8e+05	7.8e+03 /4.2e+04	1.4e+04 /4.1e+04	4.1e+06/ 1.8e+06
autorefineMaxLevel	F	2.5e+08 /1.4e+09	6.8e+08 /1.5e+09	1.3e+06 /2.0e+06	1.0e+02 /1.7e+03	2.0e+03 /3.9e+04	1.0e+02 /5.5e+02	3.9e+06/ 6.8e+05
compressMass	F	2.4e+08 /1.2e+09	6.3e+08 /1.5e+09	2.1e+05 /1.3e+06	2.4e+02 /1.6e+03	1.8e+03 /3.0e+04	2.4e+02 /2.4e+02	4.1e+06/ 3.9e+05
diffPW	M	3.6e+08 /1.9e+09	1.0e+09 /3.6e+09	3.4e+05 /5.7e+06	1.4e+05 /3.6e+05	6.1e+03 /4.6e+04	7.1e+03 /2.6e+04	4.1e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.5e+08 /1.3e+09	7.2e+08 /1.6e+09	3.2e+05 /2.7e+06	1.0e+02 /1.6e+03	1.9e+03 /4.4e+04	1.0e+02 /9.6e+02	3.9e+06/ 1.0e+06
norm2Estimate	F	2.9e+08 /1.3e+09	6.7e+08 /1.6e+09	3.6e+05 /2.7e+06	2.0e+02 /2.1e+03	1.5e+03 /4.6e+04	2.0e+02 /1.1e+03	3.8e+06/ 9.7e+05
pmapCutHistogram	F	2.9e+08 /1.4e+09	6.8e+08 /1.6e+09	4.6e+05 /1.9e+06	1.7e+03/ 1.3e+03	8.5e+03 /3.7e+04	1.7e+03/ 3.8e+02	3.9e+06/ 6.8e+05
truncateTolViolations	F	2.4e+08 /1.2e+09	6.1e+08 /1.6e+09	2.3e+05 /1.3e+06	4.6e+02 /2.1e+03	3.2e+03 /2.8e+04	4.6e+02/ 4.1e+02	4.1e+06/ 3.8e+05
Total		2.3e+09/1.2e+10	6.0e+09/1.7e+10	3.6e+06/2.3e+07	3.1e+05/6.5e+05	3.3e+04/3.1e+05	2.4e+04/7.1e+04	3.2e+07/7.7e+06

Table 36: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0843 $\pm$ 0.00107	0.0966 $\pm$ 0.00015	0.0773$\pm$0.00118	0.3663 $\pm$ 0.00210	1.09 $\times$	1.15$\times$	1.25$\times$	0.26 $\times$
sum tree	F	5/5	0%	0.0299 $\pm$ 0.00003	0.0524 $\pm$ 0.00001	0.0292$\pm$0.00012	0.2507 $\pm$ 0.00119	1.02 $\times$	1.75$\times$	1.80$\times$	0.21 $\times$
Total				0.1142	0.1490	0.1065	0.6170	1.07 $\times$	1.30$\times$	1.40$\times$	0.24 $\times$
Geomean								1.06 $\times$	1.42$\times$	1.50$\times$	0.23 $\times$

Table 37: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /1.1e+09	4.3e+05 /1.3e+06	6.6e+04 /1.3e+05	4.3e+03 /1.2e+04	3.6e+03 /4.0e+03	1.5e+06/ 1.1e+06
sum tree	F	1.5e+08/ 1.5e+08	4.8e+08 /5.1e+08	5.8e+04 /1.5e+05	1.1e+02 /3.4e+02	8.3e+02 /9.0e+03	1.1e+02 /3.4e+02	1.7e+06/ 1.2e+06
Total		4.3e+08/3.7e+08	1.2e+09/1.6e+09	4.8e+05/1.5e+06	6.6e+04/1.3e+05	5.1e+03/2.0e+04	3.7e+03/4.4e+03	3.2e+06/2.3e+06

Table 38: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.4e+08 /1.7e+09	1.2e+09 /2.7e+09	1.9e+05 /2.5e+06	7.4e+04 /2.1e+05	2.9e+03 /4.5e+04	3.7e+03 /6.9e+03	1.4e+06/ 9.5e+05
sum tree	F	2.7e+08 /1.3e+09	8.3e+08 /1.3e+09	9.8e+04 /2.7e+06	1.5e+02 /6.6e+02	2.1e+03 /4.7e+04	1.5e+02 /2.6e+02	1.4e+06/ 8.7e+05
Total		6.1e+08/3.0e+09	2.0e+09/4.0e+09	2.9e+05/5.2e+06	7.4e+04/2.1e+05	5.0e+03/9.3e+04	3.8e+03/7.2e+03	2.8e+06/1.8e+06

Table 39: Per-pass performance for **Trie**, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0142 $\pm$ 0.00000	0.0262 $\pm$ 0.00001	0.0095$\pm$0.00004	0.1496 $\pm$ 0.00106	1.50$\times$	1.84$\times$	2.76$\times$	0.18 $\times$
countLazyNodes	F	4/10	60%	0.0130 $\pm$ 0.00000	0.0269 $\pm$ 0.00001	0.0064$\pm$0.00001	0.1054 $\pm$ 0.00002	2.02$\times$	2.07$\times$	4.19$\times$	0.26 $\times$
countTerminals	F	3/10	70%	0.0120 $\pm$ 0.00001	0.0261 $\pm$ 0.00001	0.0057$\pm$0.00001	0.1193 $\pm$ 0.00118	2.10$\times$	2.18$\times$	4.56$\times$	0.22 $\times$
decayTrieStats	M	10/10	0%	0.0745$\pm$0.00012	0.0812 $\pm$ 0.00008	0.0950 $\pm$ 0.00012	0.2511 $\pm$ 0.00272	0.78 $\times$	1.09 $\times$	0.85 $\times$	0.32 $\times$
resetTraversalState	M	10/10	0%	0.0750$\pm$0.00012	0.0834 $\pm$ 0.00022	0.0931 $\pm$ 0.00018	0.2538 $\pm$ 0.00021	0.81 $\times$	1.11$\times$	0.90 $\times$	0.33 $\times$
sumPrefixFreq	F	3/10	70%	0.0121 $\pm$ 0.00006	0.0253 $\pm$ 0.00016	0.0058$\pm$0.00007	0.1112 $\pm$ 0.00003	2.08$\times$	2.09$\times$	4.34$\times$	0.23 $\times$
sumSubtreeHints	F	3/10	70%	0.0121 $\pm$ 0.00009	0.0336 $\pm$ 0.00097	0.0057$\pm$0.00001	0.1112 $\pm$ 0.00002	2.11$\times$	2.76$\times$	5.84$\times$	0.30 $\times$
Total				0.2130	0.3027	0.2214	1.102	0.96 $\times$	1.42$\times$	1.37$\times$	0.27 $\times$
Geomean								1.51$\times$	1.79$\times$	2.69$\times$	0.26 $\times$

Table 40: Per-pass PAPI counters for **Trie** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.2e+07/ 4.8e+07	1.3e+08 /2.0e+08	1.5e+06/ 1.5e+05	4.8e+02/ 5.4e+01	4.8e+04/ 5.0e+03	4.8e+02/ 5.0e+01	3.7e+06/ 6.8e+05
countLazyNodes	F	6.6e+07/ 3.2e+07	1.4e+08 /1.7e+08	1.3e+06/ 2.3e+04	1.2e+02/ 7.5e+01	1.2e+04/ 3.3e+03	1.2e+02/ 7.5e+01	4.1e+06/ 5.5e+05
countTerminals	F	6.1e+07/ 2.9e+07	1.0e+08 /1.2e+08	1.4e+06/ 9.4e+03	8.1e+01/ 5.3e+01	3.0e+04/ 2.3e+03	8.6e+01/ 5.2e+01	4.0e+06/ 2.2e+05
decayTrieStats	M	1.6e+08 /2.7e+08	3.6e+08 /9.3e+08	7.7e+05 /1.4e+06	1.2e+05 /1.7e+05	6.8e+03 /1.8e+04	4.0e+03 /2.8e+04	3.1e+06/ 1.2e+06
resetTraversalState	M	1.7e+08 /2.5e+08	2.8e+08 /8.4e+08	7.0e+05 /1.2e+06	9.1e+04 /1.3e+05	8.0e+03 /1.3e+04	3.0e+03 /1.9e+04	3.1e+06/ 8.9e+05
sumPrefixFreq	F	6.1e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.4e+06/ 1.3e+04	6.7e+01/6.7e+01	2.6e+04/ 2.5e+03	6.9e+01/ 6.6e+01	4.0e+06/ 2.2e+05
sumSubtreeHints	F	6.1e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.3e+04	7.3e+01/ 6.1e+01	1.9e+04/ 2.5e+03	7.6e+01/ 6.1e+01	3.9e+06/ 2.1e+05
Total		6.5e+08/6.9e+08	1.2e+09/2.5e+09	8.6e+06/2.8e+06	2.2e+05/3.0e+05	1.5e+05/4.7e+04	7.8e+03/4.7e+04	2.6e+07/3.9e+06

Table 41: Per-pass PAPI counters for **Trie** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.3e+08 /7.6e+08	3.3e+08 /9.1e+08	1.6e+05 /1.5e+06	1.2e+02 /1.3e+03	2.0e+03 /2.8e+04	1.2e+02 /7.7e+02	2.9e+06/ 4.8e+05
countLazyNodes	F	1.4e+08 /5.4e+08	3.2e+08 /9.1e+08	5.5e+05 /1.1e+06	1.3e+02 /1.0e+03	1.8e+03 /2.2e+04	1.3e+02 /4.3e+02	2.8e+06/ 3.3e+05
countTerminals	F	1.3e+08 /6.1e+08	2.9e+08 /8.5e+08	9.1e+05/ 6.5e+05	1.2e+02 /9.2e+02	1.9e+03 /1.6e+04	1.2e+02 /1.4e+02	2.8e+06/ 1.4e+05
decayTrieStats	M	1.9e+08 /1.1e+09	5.4e+08 /2.3e+09	3.3e+05 /4.7e+06	1.1e+05 /2.3e+05	4.8e+03 /3.3e+04	3.9e+03 /5.1e+04	2.8e+06/ 1.2e+06
resetTraversalState	M	2.1e+08 /1.1e+09	4.7e+08 /2.2e+09	4.0e+05 /3.5e+06	1.3e+05 /2.1e+05	4.5e+03 /2.8e+04	3.9e+03 /2.6e+04	2.8e+06/ 9.0e+05
sumPrefixFreq	F	1.3e+08 /5.6e+08	3.1e+08 /8.5e+08	2.0e+05 /6.1e+05	9.0e+01 /5.4e+02	1.3e+03 /1.6e+04	9.0e+01 /1.5e+02	3.0e+06/ 1.4e+05
sumSubtreeHints	F	1.7e+08 /5.6e+08	3.1e+08 /8.5e+08	4.1e+06/ 6.0e+05	1.0e+02 /6.2e+02	7.7e+04/ 1.6e+04	1.0e+02 /1.7e+02	2.4e+06/ 1.5e+05
Total		1.1e+09/5.2e+09	2.6e+09/8.9e+09	6.7e+06/1.3e+07	2.4e+05/4.4e+05	9.3e+04/1.6e+05	8.4e+03/7.9e+04	1.9e+07/3.3e+06